SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Escape Speed, Escape Energy & Orbital Trajectories

PRACTICE SHEET – 1

Escape Speed, Escape Energy and Orbital Trajectories

Name: _____	Class / Section: _____	Date: _____
Time Allowed: 60 Mins	**Total Questions: 15**	Roll No.: _____

Instructions: Attempt all questions. Section A has only one correct option; Section B needs full working. Time Allowed: 60 Mins.

Section A – Single Correct Option (Q.1 to Q.12)

Q.1 A planet in a distant solar system is 10 times more massive than the earth and its radius is 10 times smaller. Given that the escape velocity from the earth is 11 km s^{-1}, the escape velocity from the surface of the planet would be

(1) 11 km s^{-1} (2) 110 km s^{-1}
(3) 0.11 km s^{-1} (4) 1.1 km s^{-1}

[Nucleus, JM Q.1 [AIEEE-2008]]

Q.2 The mass of a spaceship is 1000 kg. It is to be launched from the earth's surface out into free space. The value of 'g' and 'R' (radius of earth) are 10 m/s^2 and 6400 km respectively. The required energy for this work will be :

(1) 6.4×10^{11} Joules (2) 6.4×10^{8} Joules
(3) 6.4×10^{9} Joules (4) 6.4×10^{10} Joules

[Nucleus, JM Q.5 [AIEEE 2012]]

Q.3 A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V. Due to the rotation of planet about its axis the acceleration due to gravity g at equator is 1/2 of g at poles. The escape velocity of a particle on the pole of planet in terms of V is

(A) $V_e = 2V$ (B) $V_e = V$
(C) $V_e = \sqrt{2}\,V$ (D) $V_e = \sqrt{3}\,V$

[Nucleus, O-1 Q.10]

Q.4 The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(A) v_e (B) $\dfrac{v_e}{\sqrt{2}}$ (C) $\dfrac{v_e}{2}$ (D) zero

[Nucleus, O-1 Q.12]

Q.5 Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of the escape velocity $\dfrac{v_A}{v_B}$ is

(A) 2 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) 1/2

[Nucleus, O-1 Q.13]

Q.6 A satellite is revolving in a circular orbit at a height 'h' from the earth's surface (radius of earth R; $h \ll R$). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to: (Neglect the effect of atmosphere.)

(1) $\sqrt{gR}$ (2) $\sqrt{gR/2}$
(3) $\sqrt{gR}\,(\sqrt{2}-1)$ (4) $\sqrt{2gR}$

[Nucleus, JM Q.9 [JEE Main 2016]]

Q.7 A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon.

(1) $\dfrac{E}{4}$ (2) $\dfrac{E}{16}$ (3) $\dfrac{E}{64}$ (4) $\dfrac{E}{32}$

[Nucleus, JM Q.15 [JEE Main-2019]]

Q.8 What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of $2R$?

(1) $\dfrac{5GmM}{6R}$ (2) $\dfrac{2GmM}{3R}$
(3) $\dfrac{GmM}{2R}$ (4) $\dfrac{GmM}{3R}$

[Nucleus, JM Q.6 [JEE Main 2013]]

Q.9 A satellite of mass m, initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is

(A) $\dfrac{\sqrt{3}}{4}mgR$ (B) $\dfrac{1}{2}mgR$
(C) $\dfrac{1}{4}mgR$ (D) $\dfrac{3}{4}mgR$

[Nucleus, O-1 Q.16]

Q.10 A satellite of the earth is revolving in circular orbit with a uniform velocity V. If the gravitational force suddenly disappears, the satellite will

(A) continue to move with the same velocity in the same orbit.
(B) move tangentially to the original orbit with velocity V.
(C) fall down with increasing velocity.
(D) come to a stop somewhere in its original orbit.

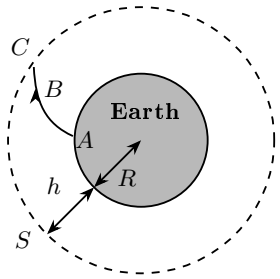
[Nucleus, O-1 Q.15]

Q.11 A satellite of mass M is in a circular orbit of radius R about the centre of the earth. A meteorite of the same mass, falling towards the earth, collides with the satellite completely inelastically. The speeds of the satellite and the meteorite are the same, just before the collision. The subsequent motion of the combined body will be :

(1) in a circular orbit of a different radius
(2) in an elliptical orbit
(3) in the same circular orbit of radius R
(4) such that it escapes to infinity

[Nucleus, JM Q.12 [JEE Main-2019]]

Q.12 *An Earth satellite is a body placed in a stable orbit about the Earth. These satellites are used for communications, meteorology, Earth measurements (gravitation and magnetic fields), resource evaluation (water, minerals), transmission of radio and TV signals, and as reference points for navigation. Although most satellites are launched from ground based stations, more recently some have been placed in orbit from one of NASA's space shuttles.*
There are two requirements needed to place a satellite in a stable orbit at an insertion point C. (Figure). It is first necessary to bring the satellite to that altitude and then the satellite must be given the necessary orbiting velocity. The orbiting velocity for a circular orbit, also called the insertion velocity.



If velocity at insertion point is k times escape velocity of the satellite at the insertion height h, then mark **incorrect** statement.

(A) For $k = 1$ satellite moves in a circular orbit
(B) For $k \geq 1$ satellite moves in an unbound orbit
(C) For $k < 1$ satellite may move in an elliptical orbit
(D) For $k < 1$ satellite can move in circular orbit

[Nucleus, O-2 Q.16 (Paragraph for Q.15 to 17)]

Section B – Subjective Problems (Q.13 to Q.15)

Q.13 A particle is fired vertically from the surface of the earth with a velocity kv_e, where v_e is the escape velocity and $k < 1$. Neglecting air resistance and assuming earth's radius as R_e. Calculate the height to which it will rise from the surface of the earth.

[Nucleus, S-1 Q.7]

Q.14 A hypothetical spherical planet of radius R and its density varies as $\rho = Kr$, where K is constant and r is the distance from the center. Determine the pressure caused by gravitational pull inside ($r < R$) the planet at a distance r measured from its center.

[Nucleus, S-1 Q.13]

Q.15 The minimum and maximum distances of a satellite from the centre of the Earth are $2R$ and $4R$ respectively, where R is the radius of Earth and M is the mass of the Earth. Find radius of curvature at the point of minimum distance.

[Nucleus, S-1 Q.14]

ANSWER KEY – Practice Sheet 1

Answers are taken from the source answer keys (Nucleus: Answer Key page, Exercises S-1, O-1, O-2, JM). JM answers are option numbers.

1. [Nucleus JM Q.1] (2)

2. [Nucleus JM Q.5] (4)

3. [Nucleus O-1 Q.10] A

4. [Nucleus O-1 Q.12] B

5. [Nucleus O-1 Q.13] A

6. [Nucleus JM Q.9] (3)

7. [Nucleus JM Q.15] (2)

8. [Nucleus JM Q.6] (1)

9. [Nucleus O-1 Q.16] D

10. [Nucleus O-1 Q.15] B

11. [Nucleus JM Q.12] (2)

12. [Nucleus O-2 Q.16] A

13. [Nucleus S-1 Q.7] $h = \dfrac{R_e\, k^2}{1 - k^2}$

14. [Nucleus S-1 Q.13] $\dfrac{1}{4} K^2 G\pi \left(R^4 - r^4\right)$

15. [Nucleus S-1 Q.14] $\dfrac{8R}{3}$

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SELF ASSESSMENT – Practice Sheet 1

Fill this in after checking your answers with the Answer Key.

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Start time: _____	End time: _____
Finished within the allowed time? ☐ Yes ☐ No	(extra / saved time: _____ Mins)

Total	Attempted	Correct	Incorrect	Not attempted
15				

Score (Correct / 15): _____ Percentage: _____ %

Question-wise tracker write ✓ (correct), ✗ (incorrect) or – (not attempted)

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	

Reason for the incorrect / unattempted questions (write question numbers)

Concept not clear: _____	Formula forgotten: _____
Calculation / silly mistake: _____	Ran out of time: _____

Personal feedback guide (tick the band that matches your score)

	Correct	What to do next
☐	14–15	Excellent. Concept and speed are both strong. Move to harder JEE-level problems and keep revising formulae once a week.
☐	11–13	Good. Re-do every incorrect question without looking at the solution, then note the formula or idea you missed.
☐	7–10	Fair. Revise the theory of escape speed, orbital speed, energies of a satellite and Kepler's laws, then re-attempt this sheet after 3 days.
☐	0–6	Needs attention. Re-read the chapter notes and solved examples first, then attempt this sheet again in two parts of 30 mins.

My personal feedback / action plan:

_____

_____

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____